

## Short Note

# Allyl Syringate

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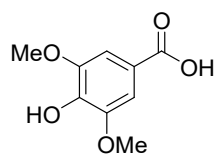
## Abstract

Syringic acid (**1**) is a naturally occurring compound with diverse biological activities. Structural modification of syringic acid often enhances its utility; however, the spectroscopic data for several derivatives remain unreported, and the procedures to obtain them can be synthetically challenging. In this study, we report the allylation of syringic acid (**1**) to afford a novel derivative, allyl syringate (**1c**). The structure of this compound was confirmed by IR, NMR, and MS spectroscopy and compared with those of closely related derivatives (**1a** and **1b**). These findings provide a useful foundation for further studies on the allylation of syringic acid and related phenolic acid derivatives.

**Keywords:** allylation; syringic acid; allyl bromide; allyl syringate

## 1. Introduction

Syringic acid (**1**), or 4-hydroxy-3,5-dimethoxybenzoic acid (Figure 1), is a naturally occurring phenolic compound present in a variety of edible plants and fruits [1,2]. It exhibits diverse biological activities, which are attributed to key functional groups—namely hydroxyl (OH), methoxy (OMe), and carboxylic acid (COOH)—attached to the aromatic ring [3]. Numerous studies have shown that derivatization of syringic acid and its analogues often yields compounds with enhanced biological activity, thereby facilitating broader investigation of their pharmacological properties [2,4–7]. However, access to such derivatives is often restricted by the synthetic challenges involved, which remain a major obstacle in the development of new bioactive molecules [8–11].



Syringic acid (**1**)

**Figure 1.** Chemical structure of syringic acid (**1**).

Allylated compounds, which are derivatives formed by introducing an allyl group into a molecule, have recently emerged as a major focus of scientific research owing to their versatility in organic synthesis and potential for diverse applications [12–19]. Their distinctive reactivity—particularly their capacity to undergo various allylic alkylation reactions—renders them valuable building blocks for the construction of complex molecules, including natural products and pharmaceuticals.



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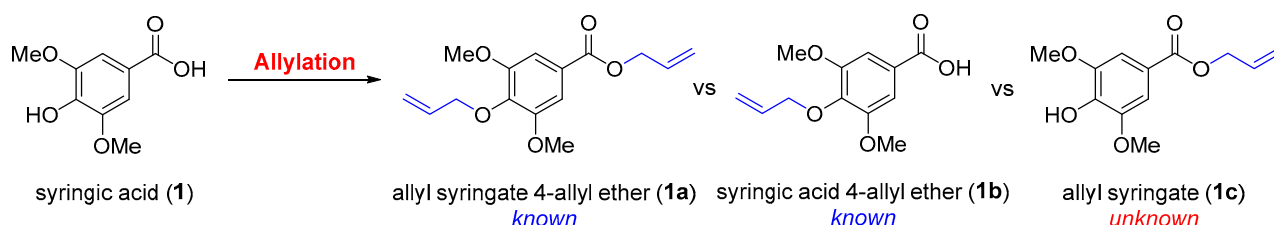
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In general, allyl groups can be introduced into organic molecules through various strategies, including allylic substitution reactions, allylation with organometallic reagents, and C–H activation [15,20–23]. These approaches exploit the unique reactivity of the allyl moiety, enabling diverse transformations and the construction of complex molecular architectures. Although numerous methods for the allylation of organic compounds have been developed, several limitations remain in achieving efficient and highly selective reactions. Common challenges include restricted substrate scope, difficulties in controlling stereochemistry, and practical constraints in large-scale applications [24,25]. Another key limitation in chemical synthesis, particularly in certain countries, is the limited availability of specific reagents [11,26,27]. This can hinder the development of new molecules and the evaluation of their properties, ultimately slowing research and innovation across multiple fields. Reactions involving complex starting materials that contain multiple functional groups further increase the likelihood of undesired side reactions [28,29]. Selective modification of a single functional group can be challenging and minimizing byproduct formation remains a critical objective. Additionally, some allylation protocols rely on toxic reagents or harsh conditions, raising environmental concerns and restricting their broader applicability [30–32]. Thus, reducing reagent toxicity and employing milder reaction conditions are essential principles in advancing sustainable synthetic methodologies [33].

In the allylation of syringic acid (**1**), two allylated derivatives (**1a–1b**) have previously been prepared in reasonable yields using conventional procedures [34–37]. However, allyl syringate (**1c**) has not yet been reported (Scheme 1). Herein, we describe the *O*-allylation of syringic acid (**1**) using a practical and convenient method. Because allyl alcohol is not readily available in Thailand, allyl bromide—a commonly accessible reagent—was employed as the allylating agent. The *O*-benzoate group was successfully allylated, affording the novel derivative, allyl syringate (**1c**), in high yield. The structure of this compound was confirmed by spectroscopic characterization and comparison with the known derivatives (**1a–1b**). These findings provide useful information for future studies on the allylation of syringic acid (**1**) and structurally related phenolic acid derivatives.

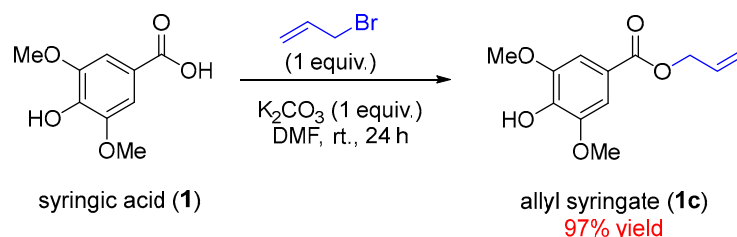


**Scheme 1.** Possible *O*-allylation products of syringic acid (**1**).

## 2. Results and Discussion

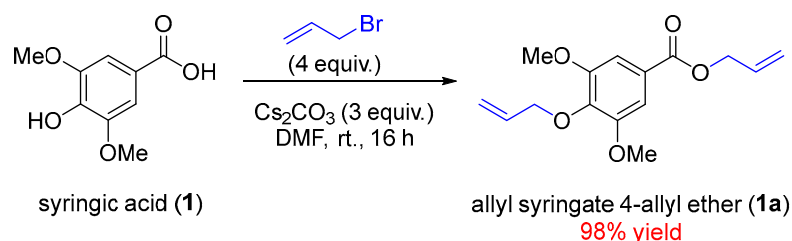
### 2.1. The Synthesis of Allyl Syringate (**1c**) and Related Derivatives (**1a–1b**)

Syringic acid (**1**) contains two reactive hydroxyl groups; therefore, three allylated products can potentially be obtained: allyl syringate 4-allyl ether (**1a**), syringic acid 4-allyl ether (**1b**), and allyl syringate (**1c**), as mentioned earlier (Scheme 1). Our work focused on the synthesis of allyl syringate (**1c**) using a commonly reported base-promoted allylation procedure [38]. Allyl bromide was chosen as the allylating agent because it is commercially available and compatible with mild base conditions, although it is not considered a “green” reagent for ideal synthesis. A mixture of syringic acid (**1**) (1 equiv.), potassium carbonate (1 equiv.), and allyl bromide (1 equiv.) in anhydrous DMF was stirred at room temperature. After stirring the reaction for 24 h, monitored the progress of the reaction by TLC, and then purification by column chromatography, allyl syringate (**1c**) was afforded in 97% yield (Scheme 2).



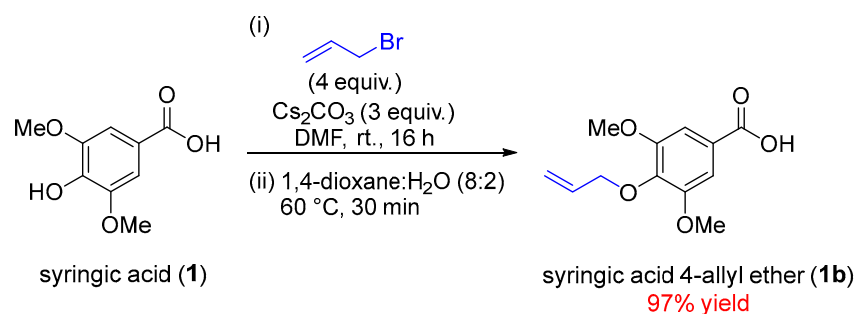
**Scheme 2.** Synthesis of allyl syringate (**1c**) from syringic acid (**1**).

As mentioned above, *O*-allylation of syringic acid (**1**) can afford three possible derivatives (**1a–1c**). To enable their unambiguous differentiation, all three compounds were synthesized, and their distinct spectroscopic characteristics were obtained. Allyl syringate 4-allyl ether (**1a**) was successfully prepared in excellent yield by a slight modification of a previously reported procedure [37]. Specifically, a mixture of syringic acid (**1**, 1 equiv.), cesium carbonate ( $\text{Cs}_2\text{CO}_3$ , 3 equiv.), and allyl bromide (4 equiv.) in anhydrous DMF was stirred at room temperature for 16 h. Subsequent purification by column chromatography afforded compound **1a** in 98% yield (Scheme 3).



**Scheme 3.** Synthesis of allyl syringate 4-allyl ether (**1a**) from syringic acid (**1**).

In the case of syringic acid 4-allyl ether (**1b**), the compound was obtained in high yield by a slight modification of a previously reported two-step procedure [37,39]. The one-pot sequence consisted of double allylation followed by hydrolysis of the allyl ester. Thus, a mixture of syringic acid (**1**, 1 equiv.), cesium carbonate ( $\text{Cs}_2\text{CO}_3$ , 3 equiv.), and allyl bromide (4 equiv.) in anhydrous DMF was stirred at room temperature for 16 h to afford crude allyl syringate 4-allyl ether (**1a**). Subsequent hydrolysis of the allyl ester group at 60 °C in a 1,4-dioxane/water (8:2) mixture furnished crude **1b**. Purification by column chromatography gave syringic acid 4-allyl ether (**1b**) in 97% yield (Scheme 4).



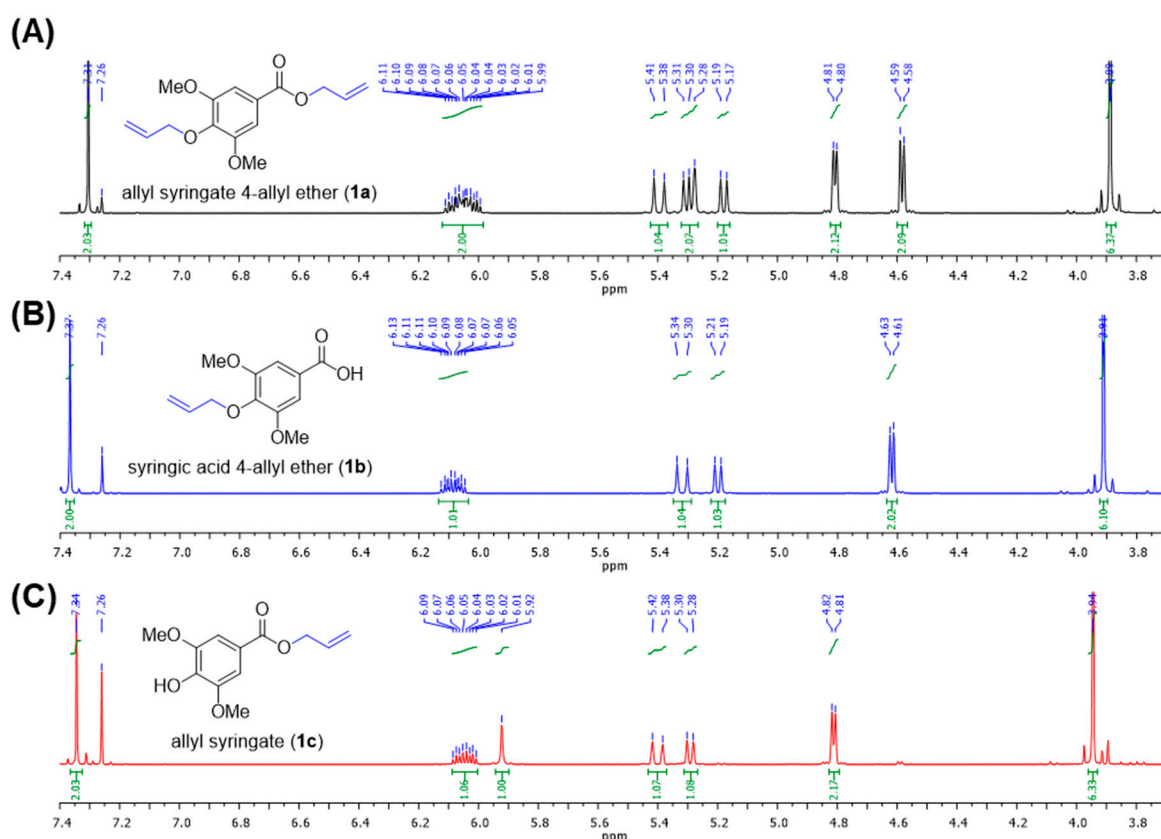
**Scheme 4.** Synthesis of syringic acid 4-allyl ether (**1b**) from syringic acid (**1**).

## 2.2. Characterization of the Allylated Compounds (**1a–1c**)

In this study, the structure of the novel compound allyl syringate (**1c**) was confirmed and characterized using spectroscopic techniques, including IR, NMR, and MS, and the results were compared with those of the related derivatives (**1a–1b**) [34–37]. In the IR spectrum (see Supporting Information), allyl syringate (**1c**) exhibited a broad absorption

band at  $3408\text{ cm}^{-1}$ , corresponding to the O–H stretching of the phenolic group, and a strong peak at  $1707\text{ cm}^{-1}$ , attributed to the C=O stretching of the ester. For syringic acid 4-allyl ether (**1b**), similar peaks were observed, together with an additional broad absorption band in the range of  $3250\text{--}2750\text{ cm}^{-1}$ , corresponding to the O–H stretching of the carboxylic acid group, and a strong peak at  $1682\text{ cm}^{-1}$ , assigned to the C=O stretching of the carboxylic acid group. In contrast, allyl syringate 4-allyl ether (**1a**) showed no absorption attributable to O–H stretching. Its major characteristic peak was the ester C=O stretching band, appearing at  $1715\text{ cm}^{-1}$ .

In the  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectra, allyl syringate (**1c**) exhibited signals in both the aromatic region and the methoxy (OMe) substituents comparable to those observed for compounds **1a** and **1b** (Figure 2). In the  $^{13}\text{C}$  NMR spectra, compounds **1a** and **1c**, both containing an ester functional group, showed characteristic carbonyl signals at 165.9 and 166.0 ppm, respectively, whereas compound **1b**, bearing a carboxylic acid group, displayed its carbonyl signal further downfield at 171.4 ppm. A comparison of the  $^1\text{H}$  NMR spectra of **1b** and **1c** revealed a distinct difference in the chemical shift of the allylic methylene group: in **1b**, the signal appeared at 4.62 ppm, while in **1c** the corresponding allylic ether methylene signal was less shielded, appearing at 4.81 ppm (Figure 2B,C). For allyl syringate 4-allyl ether (**1a**), which contains both an allyl ester and an allyl ether functionality, the  $^1\text{H}$  NMR spectrum displayed a combination of the characteristic signals of **1b** and **1c** (Figure 2A). Specifically, the methylene protons of the allyl groups in **1c** resonated at 4.58 and 4.81 ppm, respectively.



**Figure 2.**  $^1\text{H}$  NMR Spectra of compounds **1a** (A), **1b** (B), and **1c** (C).

In the mass spectrometry (MS) analysis (see Supporting Information), all three compounds (**1a–1c**) exhibited molecular ion peaks in good agreement with their calculated molecular weights, thereby supporting the proposed molecular structures.

### 3. Materials and Methods

#### 3.1. Chemicals and Instruments

Syringic acid (**1**) and anhydrous dimethyl formamide were purchased from Sigma-Aldrich (St. Louis, MO, USA). Allyl bromide was obtained from and Tokyo Chemical Industry (TCI, Tokyo, Japan). Melting points were determined using a Gallenkamp melting point apparatus (Loughborough, UK), and infrared (IR) spectra were recorded on a PerkinElmer 8900 spectrometer (Shelton, CT, USA) at the Department of Chemistry, School of Science, KMITL.  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectra were acquired on a JEOL JNM-ECZ-500R/S1 (500 MHz) spectrometer (Tokyo, Japan) at the Scientific Instruments Centre, School of Science, KMITL, with residual protonated chloroform ( $\text{CDCl}_3$ ,  $\delta$  7.26 ppm for  $^1\text{H}$  NMR and  $\delta$  77.00 ppm for  $^{13}\text{C}$  NMR) as internal standards. High-resolution mass spectra (HRMS) were measured using an Agilent 1260 Infinity Series instrument (Waldbronn, Germany) at the Faculty of Science, Naresuan University.

#### 3.2. Synthesis of 4-(Allyloxy)-3,5-dimethoxybenzoate or Allyl Syringate 4-Allyl Ether (**1a**)

In a Schlenk tube, syringic acid (**1**) (59.5 mg, 0.30 mmol), allyl bromide (145.2 mg, 1.20 mmol), and cesium carbonate (293.2 mg, 0.90 mmol) were combined, and anhydrous dimethylformamide (3 mL) was added. The reaction mixture was stirred at room temperature for 16 h. Upon completion, water (30 mL) was added, and the mixture was extracted with dichloromethane ( $3 \times 15$  mL). The combined organic layers were concentrated under reduced pressure, and the crude product was purified by flash column chromatography to afford the corresponding pure product as a colorless oil (98%).  $R_f$  = 0.83 (30% EtOAc/hexane); IR (film) 2940, 1715 (C=O), 1589, 1499, 1460, 1414, 1331, 1213, 1182, 1124, 986, 928  $\text{cm}^{-1}$ ;  $^1\text{H}$  NMR (500 MHz,  $\text{CDCl}_3$ )  $\delta$  7.31 (2H, s, ArH), 6.12–5.98 (2H, m,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 5.40 (1H, d,  $J$  = 13.8 Hz,  $\text{COOCH}_2\text{CH}=\text{CH}_2$ ), 5.32–5.27 (2H, m,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 5.18 (1H, d,  $J$  = 8.3 Hz,  $\text{PhOCH}_2\text{CH}=\text{CH}_2$ ), 4.81 (2H, d,  $J$  = 4.4 Hz,  $\text{COOCH}_2\text{CH}=\text{CH}_2$ ), 4.58 (2H, d,  $J$  = 4.9 Hz,  $\text{PhOCH}_2\text{CH}=\text{CH}_2$ ), 3.89 (6H, s,  $2 \times \text{OCH}_3$ ).  $^{13}\text{C}$  NMR (125.8 MHz,  $\text{CDCl}_3$ )  $\delta$  165.9, 153.1, 140.8, 134.0, 132.2, 125.1, 118.3, 118.2, 106.7, 74.1, 65.7, 56.2 ( $2 \times \text{CH}_3$ ). HRMS (ESI) Exact mass calcd for  $\text{C}_{15}\text{H}_{19}\text{O}_5$  [ $\text{M} + \text{H}$ ] $^+$ : 279.1233, found 279.1232. The obtained spectroscopic data were consistent with reported literature values, further confirming the proposed structure [35,36].

#### 3.3. Synthesis of 4-(Allyloxy)-3,5-dimethoxybenzoic Acid or Syringic Acid 4-Allyl Ether (**1b**)

In a Schlenk tube, syringic acid (**1**) (59.5 mg, 0.30 mmol), allyl bromide (145.2 mg, 1.20 mmol), and cesium carbonate (293.2 mg, 0.90 mmol) were combined, and anhydrous dimethylformamide (3 mL) was added. The reaction mixture was stirred at room temperature for 16 h. After complete conversion of syringic acid (**1**) to compound **1a**, as confirmed by TLC, sodium hydroxide (24 mg, 0.60 mmol), 1,4-dioxane (0.8 mL), and water (0.2 mL) were added, and the mixture was refluxed at 60 °C for 30 min. Upon completion, the pH was adjusted to neutral with 1 N HCl, water (30 mL) was added, and the mixture was extracted with dichloromethane ( $3 \times 15$  mL). The combined organic layers were concentrated under reduced pressure, and the crude product was purified by flash column chromatography to afford the pure product as a white solid (97%). m.p. 109–110 °C;  $R_f$  = 0.23 (30% EtOAc/hexane); IR (film) 3250–2750 (br, OH), 2941, 1682 (C=O), 1587, 1462, 1454, 1416, 1333, 1229, 1128, 984, 768  $\text{cm}^{-1}$ ;  $^1\text{H}$  NMR (500 MHz,  $\text{CDCl}_3$ )  $\delta$  7.37 (2H, s, ArH), 6.09 (1H, ddt,  $J$  = 13.2, 8.3, 4.9 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 5.32 (1H, d,  $J$  = 13.8 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 5.20 (1H, d,  $J$  = 8.3 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 4.62 (2H, d,  $J$  = 4.9 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 3.91 (6H, s,  $2 \times \text{OCH}_3$ ).  $^{13}\text{C}$  NMR (125.8 MHz,  $\text{CDCl}_3$ )  $\delta$  171.4, 153.2, 141.6, 134.0, 124.1, 118.3, 107.4, 56.2 ( $2 \times \text{CH}_3$ ). Exact mass calcd for  $\text{C}_{12}\text{H}_{15}\text{O}_5$  [ $\text{M} + \text{H}$ ] $^+$ : 239.0920, found

239.0922. The obtained spectroscopic data were consistent with reported literature values, further confirming the proposed structure [37].

### 3.4. Synthesis of Allyl Syringate (1c)

In a Schlenk tube, syringic acid (**1**) (59.5 mg, 0.30 mmol), allyl bromide (36.3 mg, 0.30 mmol), and potassium carbonate (41.5 mg, 0.30 mmol) were combined, and anhydrous dimethylformamide (3 mL) was added. The reaction mixture was stirred at room temperature for 24 h. Upon completion, water (30 mL) was added, and the mixture was extracted with dichloromethane ( $3 \times 15$  mL). The combined organic layers were concentrated under reduced pressure, and the crude product was purified by flash column chromatography to afford the corresponding pure product as a colorless oil (97%).  $R_f$  = 0.51 (30% EtOAc/hexane); IR (film) 3408 (brs, OH), 2938, 1707 (C=O), 1611, 1514, 1460, 1423, 1364, 1331, 1275, 1209, 1184, 1111, 988, 764  $\text{cm}^{-1}$ ;  $^1\text{H}$  NMR (500 MHz,  $\text{CDCl}_3$ )  $\delta$  7.34 (2H, s, ArH), 6.05 (1H, ddd,  $J$  = 17.9, 8.3, 4.5 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 5.92 (1H, brs, OH), 5.40 (1H, d,  $J$  = 13.7 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 5.29 (1H, d,  $J$  = 8.3 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 4.81 (2H, d,  $J$  = 4.5 Hz,  $\text{OCH}_2\text{CH}=\text{CH}_2$ ), 3.94 (6H, s,  $2 \times \text{OCH}_3$ ).  $^{13}\text{C}$  NMR (125.8 MHz,  $\text{CDCl}_3$ )  $\delta$  166.0, 146.6, 139.3, 132.4, 121.1, 118.2, 106.7, 65.6, 56.4 ( $2 \times \text{CH}_3$ ). HRMS (ESI) Exact mass calcd for  $\text{C}_{12}\text{H}_{15}\text{O}_5$   $[\text{M} + \text{H}]^+$ : 239.0920, found 239.0919.

## 4. Conclusions

In this study, we report, for the first time, the *O*-allylation of syringic acid (**1**), affording the novel compound allyl syringate (**1c**) via a practical and convenient procedure. Since allyl alcohol is not readily available in Thailand, allyl bromide—a widely accessible reagent—was employed as the allylating agent. The spectroscopic characterization (IR, NMR, and MS) has been discussed in detail and compared with the known derivatives (**1a–1b**). These findings provide a solid foundation for future studies on the allylation and alkylation of syringic acid and structurally related phenolic acid analogues.

**Supplementary Materials:** Figure S1:  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectra of allyl 4-(allyloxy)-3,5-dimethoxybenzoate **1a**; Figure S2: IR and HR-MS spectra of 4-(allyloxy)-3,5-dimethoxybenzoate **1a**; Figure S3:  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectra of 4-(allyloxy)-3,5-dimethoxybenzoic acid **1b**; Figure S4: IR and HR-MS spectra of 4-(allyloxy)-3,5-dimethoxybenzoic acid **1b**; Figure S5:  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectra of allyl syringate **1c**; Figure S6: IR and HR-MS spectra of allyl syringate **1c**.

**Author Contributions:** Conceptualization, supervision, project administration, funding acquisition, writing—original draft preparation, and writing—review and editing, N.C.; methodology, software, validation, formal analysis, investigation, resources, data curation, and visualization, N.T., S.P., L.S. and N.C. All authors have read and agreed to the published version of the manuscript.

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**Data Availability Statement:** The data supporting the findings of this study are provided in the Supplementary File.

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**Conflicts of Interest:** The authors declare no conflicts of interest.

## Abbreviations

The following abbreviations are used in this manuscript:

|        |                  |
|--------|------------------|
| rt     | Room temperature |
| equiv. | Equivalent       |
| h      | Hour             |
| min    | Minute           |
| N      | Normal           |

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